

AN UPPER BOUND FOR THE AMBIENT JUNG CONSTANT OF ℓ_1^{4k+1}

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ABSTRACT. For every integer $k \geq 1$, we prove that each nonempty bounded set in real ℓ_1^{4k+1} has an ambient enclosing ball of radius at most $(4k-1)/(4k)$ times its diameter. The bound is sharp in dimension five: $J(\ell_1^5) = 3/4$. If a Hadamard matrix of order $4k$ exists, then $J(\ell_1^{4k-1}) = J(\ell_1^{4k}) = J(\ell_1^{4k+1}) = (4k-1)/(4k)$. In particular, for every integer $r \geq 2$, $J(\ell_1^{2^r-1}) = J(\ell_1^{2^r}) = J(\ell_1^{2^r+1}) = 1 - 2^{-r}$.

1. INTRODUCTION

For a nonempty bounded set $A \subset \mathbb{R}^d$, put

$$\|x\|_1 = \sum_{j=1}^d |x_j|, \quad \text{diam } A = \sup_{x,y \in A} \|x - y\|_1, \quad \text{rad } A = \inf_{c \in \mathbb{R}^d} \sup_{x \in A} \|x - c\|_1.$$

The Jung constant is

$$J(\ell_1^d) = \sup_{\substack{A \neq \emptyset \text{ bounded} \\ \text{diam } A > 0}} \frac{\text{rad } A}{\text{diam } A}.$$

The center ranges over the whole ambient space. Using twice the radius in the definition doubles the values of the Jung constant.

Theorem 1.1. *Let $k \geq 1$ be an integer, $D \geq 0$, and let $A \subset \mathbb{R}^{4k+1}$ be nonempty. Suppose that*

$$\sum_{j=1}^{4k+1} |x_j - y_j| \leq D \quad (x, y \in A).$$

There exists $c \in \mathbb{R}^{4k+1}$ such that

$$\sum_{j=1}^{4k+1} |x_j - c_j| \leq \frac{4k-1}{4k} D \quad (x \in A).$$

Consequently $J(\ell_1^{4k+1}) \leq (4k-1)/(4k)$.

The contact reduction and coordinate-gap identities reduce the bound to a cut inequality. For $k \geq 2$, we use the quadratic-form estimate in Lemma 3.1. Section 5 gives the six-label estimate needed in dimension five and a set attaining the bound.

Corollary 1.2. *Suppose $k \geq 1$ and a Hadamard matrix of order $4k$ exists. Then*

$$J(\ell_1^{4k-1}) = J(\ell_1^{4k}) = J(\ell_1^{4k+1}) = \frac{4k-1}{4k}.$$

In particular, for every integer $r \geq 2$,

$$J(\ell_1^{2^r-1}) = J(\ell_1^{2^r}) = J(\ell_1^{2^r+1}) = 1 - 2^{-r}.$$

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The contact reduction, the dimension bound, and the lower bound obtained from Hadamard matrices are classical. Dol'nikov [2] proved the Hadamard equality criterion and the estimate $J(\ell_1^{2m}) \leq (2m-1)/(2m)$. These results give the first two equalities in Corollary 1.2. Ivanov's conjecture, as stated in the survey [1, §15], asserts that a Hadamard matrix of order $h = 4k$, $k \geq 1$, gives the value $(h-1)/h$ in dimensions $h, h+1, h+2$. Corollary 1.2 gives the conjectured value in dimension $h+1$ for $h = 4k$. The case $h+2$ and the existence of Hadamard matrices at every order divisible by four are not addressed here.

2. CONTACT REDUCTION

Lemma 2.1. *Let $F \subset \mathbb{R}^d$ be finite and nonempty, with $d \geq 1$ and $\text{diam } F \leq 1$. There is a minimizing center c with radius R . There are $q \leq d+1$, points $y_i \in F$, signs $\sigma_i \in \{-1, 1\}^d$, and positive weights α_i such that*

$$(1) \quad \sum_{i=1}^q \alpha_i = 1, \quad \sum_{i=1}^q \alpha_i \sigma_i = 0,$$

$$(2) \quad \langle \sigma_i, y_i - c \rangle = \|y_i - c\|_1 = R.$$

For every j ,

$$(3) \quad R \leq 1 - \sum_{\{i: y_i = y_j\}} \alpha_i \leq 1 - \alpha_j.$$

The selected points are allowed to repeat.

Proof. For every $x \in \mathbb{R}^d$,

$$\|x\|_1 = \max_{\sigma \in \{-1, 1\}^d} \langle \sigma, x \rangle.$$

The continuous function $f(c) = \max_{x \in F} \|x - c\|_1$ is coercive, as $f(c) \geq \|x_0 - c\|_1$ for any fixed $x_0 \in F$, and hence attains its minimum R . At a minimizing center, let S be the set of signs belonging to the active affine functions $c \mapsto \langle \sigma, x - c \rangle$. We have $0 \in \text{conv } S$. Indeed, otherwise strict separation of zero from the compact convex set $\text{conv } S$ gives a vector u with $\langle u, \sigma \rangle > 0$ for all $\sigma \in S$. Moving the center a sufficiently small positive distance in direction u then strictly decreases all active functions, while all finitely many inactive functions remain below R . This contradicts minimality.

Choose a convex representation of zero from S with minimal support. Its coefficients are positive. The chosen signs are affinely independent: otherwise a nonzero affine dependence b_i with $\sum b_i = 0$ permits replacement of α_i by $\alpha_i - tb_i$, where $t = \min_{b_i > 0} \alpha_i/b_i$, eliminating one coefficient while preserving nonnegativity and the representation. Thus $q \leq d+1$. Choose for each sign an associated active point y_i . Then $R = \langle \sigma_i, y_i - c \rangle \leq \|y_i - c\|_1 \leq R$, which proves (2). For any fixed j , balance gives

$$R = \sum_i \alpha_i \langle \sigma_i, y_i - y_j \rangle.$$

Terms with $y_i = y_j$ vanish; every other inner product is at most $\|y_i - y_j\|_1 \leq 1$. This proves (3). $\square$

Lemma 2.2. *Let $p \geq 2$, let $F \subset \ell_1^{2p-1}$ be finite and nonempty with diameter at most one, and suppose its minimum radius satisfies*

$$R > \frac{2p-3}{2p-2}.$$

The contact data in Lemma 2.1 have

$$q = 2p, \quad \sum_{i=1}^{2p} \sigma_i = 0.$$

For every $z \in \mathbb{R}^{2p-1}$,

$$(4) \quad \sum_{i=1}^{2p} \|y_i - z\|_1 \geq 2pR.$$

Proof. Equation (3) gives $\alpha_i \leq 1 - R < 1/(2p - 2)$. In each coordinate, equation (1) assigns total weight $1/2$ to each sign class. A class with at most $p - 1$ indices would have weight strictly less than $(p - 1)/(2p - 2) = 1/2$. Both classes therefore contain at least p indices. Since $q \leq 2p$, each contains exactly p indices and $q = 2p$. This proves the unweighted sign balance. Consequently

$$\sum_i \|y_i - z\|_1 \geq \sum_i \langle \sigma_i, y_i - z \rangle = \sum_i \langle \sigma_i, y_i - c \rangle + \left\langle \sum_i \sigma_i, c - z \right\rangle = 2pR.$$

No identification of the original weights with uniform weights is needed. $\square$

3. A QUADRATIC-FORM INEQUALITY FOR CUTS

Let V have $2p$ labels. Consider a finite family of cuts indexed by $\mathcal{I}$, and let S_A be a nonempty proper shore for each $A \in \mathcal{I}$. Different indices may represent the same unordered cut. Its size is $s(A) = \min(|S_A|, 2p - |S_A|)$, and its semimetric $\delta_A(i, j)$ is one when the pair crosses the cut and zero otherwise. An index is balanced when $s(A) = p$. Sums over A range over $\mathcal{I}$.

Lemma 3.1. *Let $p \geq 3$ be odd. Give each index $A \in \mathcal{I}$ a weight $w_A \geq 0$, and write*

$$d_{ij} = \sum_A w_A \delta_A(i, j), \quad \Phi = \sum_A s(A) w_A.$$

Suppose $d_{ij} \leq 1$ for all distinct i, j , and at most $2p - 1$ balanced indices have positive weight. Then

$$(5) \quad \Phi \leq B_p := \frac{p^3(2p - 1)}{p^3 + p - 1}.$$

Proof. Set

$$T = \sum_{i < j} d_{ij} = \sum_A s(A)(2p - s(A))w_A, \quad W = \sum_{s(A)=p} w_A, \quad c_p = 1 + \frac{p^2 - 1}{2p - 1}.$$

All sums over pairs are unordered sums, counting each pair once. Since an unbalanced cut has $s(A) \leq p - 1$, we have

$$(6) \quad T \geq (p + 1)\Phi - pW, \quad W \leq \frac{\Phi}{p}.$$

Also $T \leq \binom{2p}{2} = p(2p - 1)$.

We first show

$$(7) \quad T + c_p W \leq 2p^2.$$

If $W = 0$, the preceding bound on T proves this immediately. Suppose $W > 0$, and let $\mathcal{B}$ be the set of positive balanced indices, of cardinality $t \leq 2p - 1$. Choose $C \in \mathcal{B}$ with w_C maximal. Then

$$W \leq tw_C \leq (2p - 1)w_C.$$

Choose a shore S_C of size p , and put

$$v_C = \mathbf{1}_{S_C} - \mathbf{1}_{V \setminus S_C}, \quad Q_C = -\frac{1}{2}v_C^T D v_C, \quad D = (d_{ij})_{i,j \in V}.$$

Here $d_{ii} = 0$ and $\sum_i (v_C)_i = 0$. For any index A , either choice of shore gives

$$v_C^T \delta_A v_C = 2 \left(\sum_{i \in S_A} (v_C)_i \right) \left(\sum_{j \notin S_A} (v_C)_j \right) = -2 \left(\sum_{i \in S_A} (v_C)_i \right)^2.$$

It follows that

$$(8) \quad Q_C = \sum_A w_A \left(\sum_{i \in S_A} (v_C)_i \right)^2.$$

For balanced A , the inner sum is $2|S_A \cap S_C| - p$, an odd integer, so its square is at least one. For $A = C$, its square is exactly p^2 . Unbalanced indices contribute nonnegative squares. The same statements hold when another index represents the same cut as C . Using the maximal choice of w_C gives directly

$$(9) \quad Q_C \geq W + (p^2 - 1)w_C \geq \left(1 + \frac{p^2 - 1}{2p - 1}\right) W = c_p W.$$

Let X_C be the sum of d_{ij} over the pairs crossing C . Expanding the quadratic form also gives

$$Q_C = X_C - (T - X_C) = 2X_C - T \leq 2p^2 - T,$$

because there are p^2 crossing pairs, each of distance at most one. Together with (9), this gives $T + c_p W \leq T + Q_C \leq 2p^2$, proving (7).

Finally,

$$p - c_p = \frac{(p - 1)(p - 2)}{2p - 1} \geq 0.$$

Equations (6) and (7) imply

$$\begin{aligned} 2p^2 &\geq T + c_p W \\ &\geq (p + 1)\Phi - (p - c_p)W \\ &\geq \left(p + \frac{c_p}{p}\right)\Phi. \end{aligned}$$

The last step uses $W \leq \Phi/p$ and the displayed sign of $p - c_p$. Therefore

$$\Phi \leq \frac{2p^3}{p^2 + c_p} = \frac{p^3(2p - 1)}{p^3 + p - 1},$$

including the case $W = 0$. □

Remark 3.2. For every integer $p \geq 5$,

$$(10) \quad \frac{2p(2p - 3)}{2p - 2} - B_p = \frac{p(p^2 - 5p + 3)}{(p - 1)(p^3 + p - 1)} > 0.$$

Indeed $p^2 - 5p + 3 = p(p - 5) + 3 > 0$. For $p = 5$, this reads $B_5 = 375/43 < 35/4$, with difference $5/172$. For $p = 3$, the lemma gives $B_3 = 135/29 > 9/2$; the sharper six-label estimate in Section 5 is needed for the case $k = 1$.

4. COORDINATE GAPS AND PASSAGE TO ARBITRARY SETS

Lemma 4.1. *Let $y_1, \dots, y_{2p} \in \mathbb{R}^d$, with repetitions allowed. For any coordinatewise median m , there is a finite indexed family of nonnegative cut weights, with index set $\mathcal{I} = \{1, \dots, d\} \times \{1, \dots, 2p - 1\}$, such that*

$$\|y_i - y_j\|_1 = \sum_A w_A \delta_A(i, j), \quad \sum_{i=1}^{2p} \|y_i - m\|_1 = \sum_A s(A) w_A.$$

At most d balanced indices have positive weight.

Proof. In a fixed coordinate, order the values as $a_1 \leq \dots \leq a_{2p}$, retaining their labels, and put $g_j = a_{j+1} - a_j \geq 0$. At index (k, j) , assign weight g_j to the cut whose shore consists of the first j labels in coordinate k . The absolute difference for any pair is the sum of the gap weights separating its labels. Moreover, since $a_p \leq m_k \leq a_{p+1}$,

$$\sum_{i=1}^{2p} |y_{ik} - m_k| = \sum_{i=p+1}^{2p} a_i - \sum_{i=1}^p a_i = \sum_{j=1}^{2p-1} \min(j, 2p-j) g_j.$$

Summing over coordinates gives both identities without combining equal cuts. Only the index (k, p) is balanced within coordinate k , so there are at most d positive balanced indices. Ties have zero gap and may be ordered arbitrarily. If weights on equal unordered cuts are combined, the two identities remain unchanged, and the number of distinct positive balanced cuts is also at most d . $\square$

Lemma 4.2. *Let $d \geq 1$ and $\rho \geq 0$. Suppose every finite nonempty subset of ℓ_1^d of diameter at most one has an ambient enclosing ball of radius at most ρ . If a nonempty $A \subset \mathbb{R}^d$ satisfies $\|x - y\|_1 \leq D$ for every $x, y \in A$, where $D \geq 0$, then some $c \in \mathbb{R}^d$ satisfies $\|x - c\|_1 \leq \rho D$ for every $x \in A$.*

Proof. If $D = 0$, all points coincide and their common value is a suitable center. Suppose $D > 0$. Applying the finite assumption to the points of any finite nonempty $F \subset A$ divided by D , and multiplying the resulting center by D , gives an enclosing radius $r = \rho D$. For each $x \in A$ put

$$C_x = \{c \in \mathbb{R}^d : \|x - c\|_1 \leq r\}.$$

Fix $x_0 \in A$. The sets $C_x \cap C_{x_0}$ are closed subsets of the compact ball C_{x_0} . Every finite intersection is nonempty: apply the finite assertion to its indexed points together with x_0 . The finite-intersection property gives $\bigcap_{x \in A} C_x \neq \emptyset$. A point in this intersection is the required center. $\square$

5. THE FIVE-DIMENSIONAL CASE

The five-dimensional bound rests on a capacity estimate for cuts on six labels.

5.1. The six-label capacity estimate. Let $V = \{1, \dots, 6\}$. A *cut* is an unordered partition $C = \{S, V \setminus S\}$ with $\emptyset \neq S \neq V$. Its size and cut semimetric are

$$s(C) = \min\{|S|, 6 - |S|\}, \quad \delta_C(i, j) = \begin{cases} 1, & i \text{ and } j \text{ are in different shores of } C, \\ 0, & \text{otherwise.} \end{cases}$$

There are six cuts of size one, fifteen of size two, and ten of size three. We call the last ten cuts *balanced*.

Proposition 5.1. *Assign a nonnegative weight w_C to every cut of V , and put*

$$d_{ij} = \sum_C w_C \delta_C(i, j).$$

Suppose that $d_{ij} \leq 1$ for all $i \neq j$ and that at most five balanced cuts have positive weight. Then

$$(11) \quad \Phi := \sum_C s(C) w_C \leq \frac{9}{2}.$$

To describe the balanced supports, fix 6 as the base label. For a balanced cut, let T be its unique shore not containing 6. Its complement in $\{1, \dots, 5\}$ is an edge e of K_5 . Thus the ten balanced cuts correspond to the ten edges, with

$$(12) \quad T(e) = \{1, \dots, 5\} \setminus e.$$

We abbreviate the edge $\{i, j\}$ by ij .

Lemma 5.2. *Every family of at most five balanced cuts can be extended to a family of five and then relabeled by a permutation of V so that its representing edges are one of*

$$\begin{aligned} A &= \{12, 23, 34, 45, 15\}, \\ B &= \{13, 14, 23, 24, 34\}, \\ C &= \{12, 13, 23, 14, 15\}. \end{aligned}$$

Proof. There are ten balanced cuts, so a family of fewer than five can be extended. Consider its representing graph on five vertices with five edges. If disconnected, it must have a component on four vertices: components of size at most three allow at most $\binom{3}{2} + \binom{2}{2} = 4$ edges in total. It is therefore K_4 with one edge deleted, together with an isolated vertex.

If connected, the graph has a spanning tree with four edges and exactly one additional edge. It has a unique cycle, of length three, four, or five. A five-cycle uses all vertices; a four-cycle has one attached leaf. For a triangle, the two remaining vertices form a length-two tail, two leaves at one triangle vertex, or leaves at two different triangle vertices. These and the disconnected graph are all six possibilities up to permutations of $\{1, \dots, 5\}$.

It remains to use permutations involving the base. Exchanging labels 6 and $v \in \{1, \dots, 5\}$ fixes a representing edge containing v . An edge e not containing v is replaced by its complementary edge in $\{1, \dots, 5\} \setminus \{v\}$. To check this rule, apply the transposition to the two shores and then choose the shore excluding 6 in (12). If $v \notin e$, the new chosen shore is $e \cup \{v\}$, giving exactly the asserted complement.

The following three applications of this rule pair the six graph types:

Starting edges	Exchanged base label	Resulting edges
12, 23, 34, 45, 15	5	34, 14, 12, 45, 15
12, 23, 34, 14, 15	1	12, 45, 25, 14, 15
12, 23, 13, 14, 45	5	34, 14, 24, 23, 45

In the first row, a five-cycle becomes a triangle with leaves at two different vertices. In the second, a four-cycle with a leaf becomes K_4 minus an edge and an isolated vertex. In the third, a triangle with a length-two tail becomes a triangle with two leaves at one vertex. Ordinary permutations identify the representatives of these three pairs with A , B , and C , respectively. The base exchanges are involutions, so the same reductions work in either direction. $\square$

For nonnegative weights a_{ij} on the fifteen unordered pairs of V , define the capacity of a cut by

$$\text{cap}_a(S) = \sum_{\substack{i \in S \\ j \notin S}} a_{ij}.$$

This does not depend on the choice of shore. If $\deg_a(i) = \sum_{j \neq i} a_{ij}$, then

$$(13) \quad \text{cap}_a(S) = \sum_{i \in S} \deg_a(i) - 2 \sum_{\{i,j\} \subset S} a_{ij}.$$

Expanding the sum gives

$$(14) \quad \sum_{i < j} a_{ij} d_{ij} = \sum_C w_C \text{cap}_a(C).$$

Thus, if the capacity is at least one on every singleton cut, at least two on every cut of size two, and at least three on each permitted balanced cut, then nonnegativity gives

$$(15) \quad \Phi \leq \sum_{i < j} a_{ij} d_{ij} \leq \sum_{i < j} a_{ij}.$$

Balanced cuts of weight zero impose no condition in this argument.

Proof of Proposition 5.1. Extend the positive balanced support to five cuts, giving added cuts weight zero, and apply Lemma 5.2. Relabeling preserves all pairwise upper bounds, nonnegative weights, and Φ . We give a certificate of total weight at most $9/2$ in each representative.

Support A. On the ordinary vertices $1, \dots, 5$, give an edge of the five-cycle weight $2/15$ and a nonedge weight $8/15$. Give each pair $i6$ weight $1/5$. The total weight is

$$5 \frac{2}{15} + 5 \frac{8}{15} + 5 \frac{1}{5} = \frac{13}{3} < \frac{9}{2}.$$

The weighted degree of each ordinary vertex is $23/15$, and that of vertex 6 is one. Singleton capacities are therefore at least one. The size-two capacities, according as the pair is a cycle edge, an ordinary nonedge, or a pair containing the base, are

$$2 \frac{23}{15} - 2 \frac{2}{15} = \frac{14}{5}, \quad 2 \frac{23}{15} - 2 \frac{8}{15} = 2, \quad 1 + \frac{23}{15} - \frac{2}{5} = \frac{32}{15}.$$

All are at least two. For any allowed balanced shore $T(e)$, deleting the endpoints of the cycle edge e leaves a triple with two cycle edges and one ordinary nonedge. By (13), its capacity is

$$3 \frac{23}{15} - 2 \left(2 \frac{2}{15} + \frac{8}{15} \right) = 3.$$

Support B. Give all fifteen pairs weight $1/4$ and add $3/4$ to pair 56 . The total is $15/4 + 3/4 = 9/2$. Singleton capacities are at least $5/4$, and size-two capacities are at least $8/4 = 2$. Vertex 5 is isolated in the representing graph B , so every allowed shore $T(e)$ contains 5 and excludes 6. Its cut separates the pair 56 , and hence its capacity is $9/4 + 3/4 = 3$.

Support C. Start with weight $1/4$ on every pair and set

$$a_{16} = 0, \quad a_{12} = a_{13} = a_{46} = a_{56} = \frac{1}{2},$$

leaving all other weights unchanged. The total is $15/4 - 1/4 + 4/4 = 9/2$. Every vertex has weighted degree $3/2$. Consequently singleton capacities are $3/2$, and every size-two capacity is

$$3 - 2a_{ij} \geq 2.$$

The five allowed shores are

$$345, \quad 245, \quad 145, \quad 235, \quad 234,$$

where a triple denotes its set of labels. Each has all three internal pair weights equal to $1/4$. Its capacity is therefore $3(3/2) - 2(3/4) = 3$.

In all three cases, (15) proves (11). Every pair weight used is nonnegative, so the argument uses only the upper bounds $d_{ij} \leq 1$. $\square$

5.2. The radius bound and sharpness.

Theorem 5.3. *Every nonempty $A \subset \mathbb{R}^5$ whose pairwise ℓ_1 distances are at most $D \geq 0$ has an ambient enclosing ball of radius at most $3D/4$. Moreover,*

$$J(\ell_1^5) = \frac{3}{4}.$$

Proof. First take a finite nonempty set $F \subset \mathbb{R}^5$ with diameter at most one and let R be its minimum radius. If $R > 3/4$, apply Lemma 2.2 with $p = 3$ to obtain six contacts $y_i \in F$ such that for every $z \in \mathbb{R}^5$,

$$\sum_{i=1}^6 \|y_i - z\|_1 \geq 6R > \frac{9}{2}.$$

Choose a coordinatewise median m of these contacts. Apply Lemma 4.1 and combine weights on equal unordered cuts. The weights preserve all original pairwise distances and the median functional, and at most five distinct balanced cuts have positive weight. Proposition 5.1 therefore gives

$$\sum_{i=1}^6 \|y_i - m\|_1 = \Phi \leq \frac{9}{2},$$

a contradiction. Thus $R \leq 3/4$ for every finite diameter-one set. Lemma 4.2 with $\rho = 3/4$ proves the stated upper bound for all nonempty sets, including $D = 0$.

For sharpness, take the four points

$$\begin{aligned} x_1 &= \frac{1}{4}(1, 1, 1, 0, 0), & x_2 &= \frac{1}{4}(1, -1, -1, 0, 0), \\ x_3 &= \frac{1}{4}(-1, 1, -1, 0, 0), & x_4 &= \frac{1}{4}(-1, -1, 1, 0, 0). \end{aligned}$$

Every pair of distinct points differs in exactly two coordinates, so its distance is one. The zero center has distance $3/4$ from each point. In every one of the first three coordinates, two points have value $1/4$ and two have value $-1/4$. For any ambient center $c \in \mathbb{R}^5$,

$$\begin{aligned} \sum_{i=1}^4 \|x_i - c\|_1 &= 2 \sum_{j=1}^3 (|c_j - 1/4| + |c_j + 1/4|) + 4(|c_4| + |c_5|) \\ &\geq 3. \end{aligned}$$

The inequality uses $|t - 1/4| + |t + 1/4| \geq 1/2$. Hence at least one point has distance at least $3/4$ from every center. The four-point set has diameter one and ambient radius exactly $3/4$, which proves the matching lower bound. $\square$

6. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.1. For $k = 1$, the assertion is exactly the upper bound in Theorem 5.3. Suppose henceforth that $k \geq 2$ and put $p = 2k + 1$, so $p \geq 5$ is odd and $d = 2p - 1$. Let $F \subset \ell_1^d$ be finite and nonempty with diameter at most one. Let R be its minimum radius and suppose

$$R > \frac{2p - 3}{2p - 2} = \frac{4k - 1}{4k}.$$

Lemma 2.2 gives $2p$ contacts with (4). Choose a coordinatewise median m of those contacts and use Lemma 4.1. The resulting indexed cut metric has all pairwise distances at most one and at most $d = 2p - 1$ positive balanced indices. Lemma 3.1 and (10) give

$$2pR \leq \sum_i \|y_i - m\|_1 = \Phi \leq B_p < \frac{2p(2p - 3)}{2p - 2},$$

a contradiction. This proves the finite diameter-one assertion. Apply Lemma 4.2 with $\rho = (4k - 1)/(4k)$ to obtain the conclusion for the given nonempty A and every $D \geq 0$. $\square$

Remark 6.1. The bound $B_p/(2p)$ applies only to the special contact configuration forced by a radius above the threshold. It is not an unconditional upper bound for $J(\ell_1^{2p-1})$; making that substitution would contradict the Hadamard lower examples.

7. HADAMARD MATRICES AND SHARPNESS

We recall the two classical upper bounds. Lemma 2.1 has some $\alpha_j \geq 1/q \geq 1/(d + 1)$, whence $R \leq 1 - \alpha_j \leq d/(d + 1)$. For even $d = 2m$, if $R > (2m - 1)/(2m)$, every $\alpha_i < 1/(2m)$. Each sign class of weight $1/2$ would then need at least $m + 1$ indices, contradicting $q \leq 2m + 1$. Thus

$$J(\ell_1^d) \leq \frac{d}{d + 1}, \quad J(\ell_1^{2m}) \leq \frac{2m - 1}{2m}.$$

The same scaling and compactness argument extends these finite estimates to all bounded sets.

Proposition 7.1. *If a Hadamard matrix of order $h \geq 2$ exists, then for every $d \geq h - 1$ there is an h -point set in ℓ_1^d of diameter one and ambient radius $(h - 1)/h$.*

Proof. Normalize the first column of the Hadamard matrix to all ones by changing row signs. Its rows and columns are orthogonal. Delete the first column, divide each row by h , and append zeros to obtain points $x_1, \dots, x_h \in \mathbb{R}^d$. Two distinct rows differ in exactly $h/2$ entries; none of these is in the deleted column. Their distance is thus $(h/2)(2/h) = 1$. Each point has norm $(h - 1)/h$.

Every remaining Hadamard column is orthogonal to the constant column, and therefore has $h/2$ entries of each sign. For every ambient center $c \in \mathbb{R}^d$,

$$\begin{aligned} \frac{1}{h} \sum_{i=1}^h \|x_i - c\|_1 &= \frac{1}{2} \sum_{j=1}^{h-1} (|c_j - 1/h| + |c_j + 1/h|) + \sum_{j=h}^d |c_j| \\ &\geq \frac{h - 1}{h}. \end{aligned}$$

For every center, at least one point has distance at least this value. The zero center attains it for all points, proving the assertion. $\square$

Proof of Corollary 1.2. Use Proposition 7.1 with $h = 4k$ for the three lower bounds. The classical dimension bound, the even-dimensional bound, and Theorem 1.1, respectively, give the matching upper bounds for every $k \geq 1$.

For the unconditional family, start with $H_1 = (1)$ and iterate

$$H_{2h} = \begin{pmatrix} H_h & H_h \\ H_h & -H_h \end{pmatrix}.$$

If $H_h H_h^\top = hI_h$, direct block multiplication gives $H_{2h} H_{2h}^\top = 2hI_{2h}$. Thus a Hadamard matrix of order 2^r exists for every $r \geq 0$. For $r \geq 2$, set $4k = 2^r$. $\square$

The first few values in this family are

dimensions	Jung constant
3, 4, 5	3/4
7, 8, 9	7/8
15, 16, 17	15/16
31, 32, 33	31/32

The four-point set in Theorem 5.3 attains the value in dimension five.

LEAN FORMALIZATION

Lean proofs of Theorem 1.1 and the five-dimensional sharpness result are available at <https://github.com/lixiaoms/Jung>.

- `lean-toolchain` specifies the Lean version; `lakefile.lean` defines the proof libraries; `lake-manifest.json` fixes the direct and transitive dependency revisions.
- `Jung.lean` and `Jung5.lean` are the library entry points. `Jung/` contains the proof of the general bound, with the statement for every $k \geq 1$ in `Jung/Main.lean`. `Jung5/` contains the five-dimensional proof, the finite capacity certificates, and the sharpness example; its main statements are in `Jung5/Main.lean`.
- `Audit.lean` prints the types and transitive axiom dependencies of 151 declarations. The allowed axioms are `propext`, `Classical.choice`, and `Quot.sound`.
- `verify.py` checks the proof and configuration hashes, builds both libraries, and runs `Audit.lean`. It checks that every expected axiom report is present and uses only the three axioms listed above.
- The audit output is in `verification/audit.log`. `verification/source_manifest.json` records the proof and configuration hashes; `verification/summary.json` records the verified statements and versions. `release_manifest.json` lists the distributed files and their hashes.
- `BUILD.md` gives the commands for reproducing the build and audit; `PROOF_GUIDE.md` relates the Lean modules to the proof.

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